

Fig. 4: The Assembly code debugger

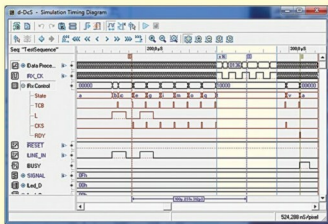


Fig. 5: The d-DcS Digital Circuit Simulator allows timing simulation of the logic networks

all the structures involved in the hardware/software system by stepping the execution of programs.

Applications of the software

Deeds was conceived as a learning environment for students interested in digital electronics. As such, it may contain different technical subjects and different pedagogical formats (lectures, exercises, lab assignments, etc) delivered at different student levels. Deeds can therefore be classified as a set of tools that teachers can complete and personalise to suit their pedagogical needs by contributing to the lecture space with their own material. A major selling point is that teachers don't need a specific authoring tool because the lecture space can be composed

the corresponding simulator, and opens that schematic together with another window (the assistant browser) that contains step-by-step instructions on how to explore or test the circuit itself.

Such a procedure is equally useful for conveying concepts of simple components or even complex networks. In the first case, the simulator allows you to explore them interactively. In the second case, signal tracing capabilities in time and data domains allow a thorough test of the network.

Solving exercises. Traditional exercises aim to help understand theory by applying it to simple cases and providing a feedback to the teacher through the delivery of solutions. In Deeds system exercises, the exercises are presented as HTML pages,

with just about any HTML editor.

Teaching lectures. A lecture based on Deeds appears as HTML pages with text and figures. The look and feel of the layout can be totally customised by the author. At this level, a student only 'sees' a normal online book or a document. But many of the figures and visual objects are active because they are connected to simulation and editing tools of Deeds.

For example, let us suppose that a theory presents a certain digital circuit, visualising its schematics in a picture. When the user clicks the picture, Deeds immediately launch-

ing the corresponding simulator, and opens that schematic together with another window (the assistant browser) that contains step-by-step instructions on how to explore or test the circuit itself.

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Now users can utilise Deeds to download the assignment from a webpage. Project development phases are guided by help and instructions supplied through the assistant browser. The instructions provided are not step-by-step and require intelligent use of simulation tools and good user initiative.

A great learning tool

Deeds is a great learning tool for all you design enthusiasts. Its flexibility, diversity and ease-of-use make it one of the best tools in the market for anyone interested in digital design. ●